GLOBE SCOTT MOTORS · FREE ENGINEERING RESOURCE

Custom DC Motor Supplier Evaluation Scorecard

For OEM procurement, engineering and quality teams sourcing custom PMDC, geared or BLDC motors for series production

Work through Steps 1 to 3 while comparing quotations, Steps 4 and 5 before releasing the series order, and Step 6 before the first delivery arrives. Score each supplier in the table at the end against evidence, not claims.

Step 1 — Before requesting quotations

- [ ] Freeze the motor specification: load torque and speed, duty cycle, voltage range at the terminals, envelope, mounting, shaft, connectors and options.
- [ ] State annual volume, call-off pattern and expected product life including service years.
- [ ] Estimate your own cost per event: incoming inspection per motor, line reject and field failure (visit, replacement, freight, downtime).
- [ ] Fix the scorecard weights before quotations arrive, so that price cannot reshape them afterwards.

Step 2 — Supplier capability evidence

- [ ] Routine test list applied to every motor, with a sample test record.
- [ ] Magnet incoming checks: flux density and coercivity, and how magnet lots are traced to motors.
- [ ] Armature tests: surge and HV dielectric test coverage (all commutator bars, every armature or a sample).
- [ ] Dynamometer load testing: every motor or a sample; parameters recorded (torque, speed, current, efficiency).
- [ ] Endurance and temperature-rise testing for the insulation class claimed (for example Class F, 155 °C).
- [ ] Quality system certificate (for example ISO 9001:2015) and audited approvals relevant to your sector.

Step 3 — Total cost of ownership worksheet (per motor)

- [ ] Unit price, stating whether ex-works or delivered.
- [ ] Incoming inspection cost: 100 % run test, or sampling supported by supplier test records.
- [ ] Line reject cost = reject rate × cost per reject.
- [ ] Field failure cost = warranty failure rate × cost per failure.
- [ ] Energy cost for the machine owner = (output power / efficiency) × hours per year × tariff, with efficiency verified at the working load point.
- [ ] Break-even check: price difference / cost per field failure = failure-rate gap that cancels the saving (example: ₹800 / ₹25,000 = 3.2 %).

Step 4 — Sample approval and golden sample

- [ ] Write acceptance criteria with nominal, tolerance and test condition: no-load speed and current, load-point speed, current and efficiency, temperature rise, insulation resistance, hi-pot, rotation, critical dimensions, backlash and noise.
- [ ] State test conditions for each: supply voltage, load point, winding temperature and instrument.
- [ ] Approve against the written criteria, not machine trials alone.
- [ ] Seal 2–3 golden samples with their test data; retain at both OEM and supplier.
- [ ] Re-verify against the golden sample after every notified change.

Step 5 — Contract clauses to agree in writing

- [ ] PCN: changes requiring notice (magnet grade, wire, brush grade, bearing, grease, gear material, sub-supplier, location), notice period and re-qualification tests.
- [ ] Warranty: start point, failure definition, exclusions and return process.
- [ ] Failure analysis: who performs it, response time, report content (root cause and corrective action).
- [ ] Ownership of customer-specific drawings and paid-for tooling; tooling location and return terms.
- [ ] Packaging: shaft protection, VCI or sealed bags with desiccant, stacking limits, lot and date code labels.
- [ ] Lifecycle: spares availability, discontinuation notice and last-time-buy window.

Step 6 — Incoming inspection plan

- [ ] Classify characteristics as critical, major or minor and set a sampling plan for each (for example from ISO 2859-1).
- [ ] Inspect dimensions, rotation, no-load speed and current, insulation resistance, visual and packaging condition.
- [ ] Remember: with n = 50 and c = 1, a lot that is 4 % defective is still accepted about 40 % of the time.
- [ ] Define the reaction to a rejected lot: 100 % screening, return, or supplier containment with a root-cause report.
- [ ] Move to reduced inspection only after a run of accepted lots backed by supplier test records.

Weighted Supplier Scorecard (score 1–5, weighted score = weight × score)

Criterion	Weight	Supplier A score	Supplier B score	Evidence seen
Testing and production consistency	0.20			
Total cost of ownership	0.20			
Engineering and customisation	0.15			
Quality system and change control	0.15			
Delivery and capacity	0.10			
Contract terms	0.10			
Track record and continuity	0.10			
Total weighted score	1.00			

Weights and cost figures are examples. Adjust them to your product, volumes and risk, and keep the completed scorecard with the supplier approval record.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.